GOVERNEMENT OF KARNATAKA
KARNATAKA SCHOOL EXAMINATION AND ASSESSMENT BOARD
6th CROSS, MALLESHWARAM, BENGALURU-560 003

II PUC CHEMISTRY

Experiments to be conducted for the Academic year 2025-26

Experiment No.	Experiments
1.	To determine the concentration / molarity of Potassium Permanganate solution by titrating it against 0.1 M standard solution of Ferrous Ammonium Sulphate.
2.	To determine the concentration / molarity of Potassium Permanganate solution by titrating it against 0.1 M standard solution of Oxalic acid
3.	Systematic qualitative analysis of simple Inorganic Salt (General procedure to be explained and recorded)
4.	Systematic qualitative analysis of simple inorganic salt 1 and 2
5.	Systematic qualitative analysis of simple inorganic salt 3 and 4
6.	Systematic qualitative analysis of simple inorganic salt 5 and 6
7.	Systematic qualitative analysis of simple inorganic salt 7 and 8
8.	Systematic qualitative analysis of simple inorganic salt 9 and 10
9.	Tests for functional groups in organic compounds
10.	Effect of concentration on rate of reaction between sodium thiosulphate and hydrochloric acid
11.	Effect of temperature on rate of reaction between sodium thiosulphate and hydrochloric acid
12.	Determination of enthalpy of solution of KNO_3 or $CuSO_4$
13.	Determination of enthalpy of neutralization of HCl and NaOH
14.	Determination of enthalpy change on mixing Chloroform and Acetone
15.	To study the variation in cell potential of the cell Zn | Zn^{2+} || Cu^{2+} | Cu with change in concentration of electrolytes ($CuSO_4$ / $ZnSO_4$) at room temperature
16.	Separation of pigments present in leaves and flowers by paper chromatography and determination of R_f values of components
17.	Separation of constituents of a mixture of inorganic components containing two cations Pb^{2+} and Cd^{2+} using chromatographic technique
18.	Preparation of Ferrous Ammonium Sulphate crystals (Mohr's salt)
19.	Preparation of potash alum
20.	Preparation of dibenzal acetone (Dibenzylidene acetone)
21.	Preparation of p-nitroacetanilide from acetanilide
22.	Preparation of phenyl azo β-naphthol (an azo dye)

Experiments to be conducted for practical examination

Time: 2 Hrs **Total Marks: 30**

Q - I	**SALT ANALYSIS** Analyse the given simple inorganic salt systematically and report one **acid radical** and one **basic radical.**	10 marks
Q - II	**TITRATION (Volumetric Analysis)** Estimate the Molarity of $KMnO_4$ solution using given standard (0.1M) **FAS** solution. (Procedure of the titration should be given).	10 marks
III	**Attendance of both theory and practical classes 75% and above**	5 marks
IV	Submission of the duly completed and certified record	5 marks
	TOTAL	**30 marks**

The following salts are suggested to be given for analysis for practical examination:

NH_4Br, NH_4Cl, $Al_2(SO_4)_3$, $MnSO_4$, $ZnSO_4$/ $ZnCO_3$

$BaCl_2$/ $Ba(NO_3)_2$, $Sr(NO_3)_2$ $CaCO_3$, $MgSO_4$/ $MgCO_3$.

SCHEME OF VALUATION

Time: 2 Hours **Total Marks: 30**

Q-I	**I. SALT ANALYSIS (10 MARKS)**			
	(i)	**Preliminary tests** (any two correct)	1 Mark	**10 Marks**
	(ii)	**Detection of Acid radical**	4 Marks	
		Correct group identification — 1 Mark		
		Correct radical identification — 1 Mark		
		Confirmatory test — 2 Marks		
	(iii)	**Detection of Basic radical**	4 Marks	
		Correct group identification — 1 Mark		
		Correct radical identification — 1 Mark		
		Confirmatory test — 2 Marks		
		For writing systematic procedure with absence of previous groups	1 Mark	

<table>
<tr><td rowspan="11">Q-II</td><td colspan="4">II. TITRATION (10 MARKS)</td></tr>
<tr><td>(i)</td><td colspan="2">For performing the experiment</td><td>3 Marks</td><td rowspan="10">10 Marks</td></tr>
<tr><td>(ii)</td><td colspan="2">For recording the readings in the tabular column</td><td>1 Mark</td></tr>
<tr><td rowspan="5">(iii)</td><td colspan="2">For accuracy of the Titre value:</td><td rowspan="5">3 Marks</td></tr>
<tr><td>Upto ± 0.3 cm^3 error</td><td>3 Marks</td></tr>
<tr><td>± 0.4 cm^3 error</td><td>2 Marks</td></tr>
<tr><td>± 0.5 cm^3</td><td>1 Mark</td></tr>
<tr><td>± 0.6 cm^3 & above</td><td>0 Mark</td></tr>
<tr><td rowspan="4">(iv)</td><td colspan="2">Calculating of Molarity:</td><td rowspan="4">3 Marks</td></tr>
<tr><td>Formula</td><td>1 Mark</td></tr>
<tr><td>Substitution</td><td>1 Mark</td></tr>
<tr><td>Final answer with unit</td><td>1 Mark</td></tr>
<tr><td>III</td><td colspan="3">Attendance of both Theory and Practical Classes 75% and above</td><td>5 Marks</td></tr>
<tr><td>IV</td><td colspan="3">Submission of the duly completed and Certified Record</td><td>5 Marks</td></tr>
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